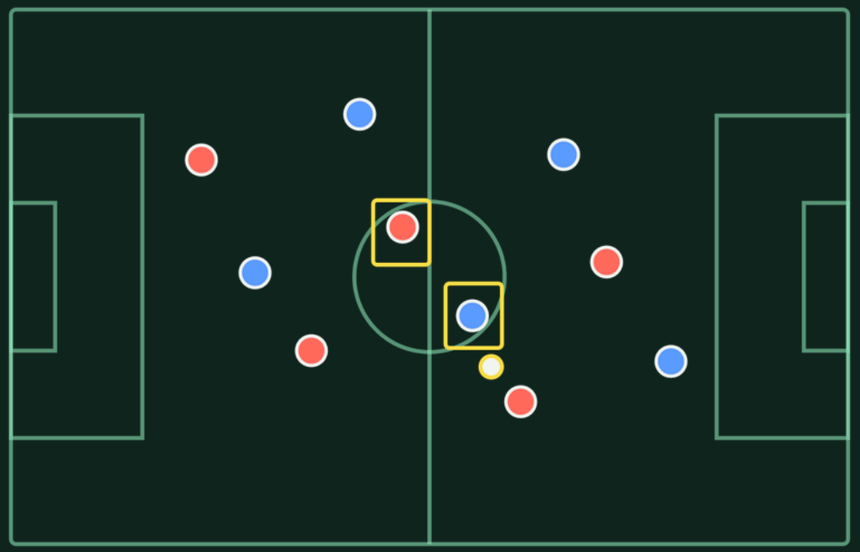
AAI3001 · COMPUTER VISION AND DEEP LEARNING

PitchProfile

Turning a football match video into player statistics



[Group 25]

Player data without the cost of collecting it

Football and data

Why football: it reaches billions. About 5 billion people engaged with the 2022 World Cup.

How statistics help: clubs use them to find players, plan matches and manage workload.

In practice: Brentford pairs data with scouting to find undervalued players.

How it improves

Today: trained operators tag events by hand. Accurate, but it takes people.
Now: found from the video.

Today: about 30 cameras per stadium, or GPS vests for one team.
Now: one video, both teams.

Today: youth and women's matches are mostly not covered.
Now: any match on video.

How it can be used

Scouting: compare players from match video.

Match review: team shape, passing and pressing.

Clubs without a data feed: academies, women's and lower leagues.

Not new: broadcast-video tracking is already sold (for example by SkillCorner), and one 2026 study found it close to stadium cameras for most events. This is an open research prototype, and for now a complement.

Sources: FIFA, Frost 2025, Rein 2016, Ravé 2020, ESPN, Liu 2013, Pappalardo 2019, Mills 2026 (full list in notes)

The open problem is the data

PUBLIC DATASET	WHAT IT GIVES	WHAT WE USED IT FOR
SoccerNet Ball Action Spotting	12 ball actions with their times	Trains and tests the action spotter
FOOTPASS (SoccerNet 2026, NDA)	Every player’s box, number and team, plus 8 action types	Trains the identity model; tests naming
SoccerNet Jersey 2023	Shirt numbers of player tracklets	Tests the number reader
SoccerNet tracking	Player and ball boxes in every frame	Trains and tests the ball detector
SoccerTrack v2	Positions, ball and every action, one match	Tunes and tests the event rules

SoccerNet, FOOTPASS and SoccerTrack are the datasets used in this project · Pappalardo et al. 2019 on provider-owned data

Why broadcast video is hard

The camera keeps cutting

The picture changes every few seconds, and only about 60% of the time shows a usable view of the pitch.

The ball is tiny

We find it in under half of the frames where it is visible.

Tackles look like touches

From positions alone they are hard to tell apart. In unseen games we found 1 of 26.

Naming every player

Only about half of player time is tied to a shirt number.

From upload to report in four phases

1 · Upload

You: add a full half or a full match video.

App: checks length, size and picture quality, and estimates the time.

2 · Analyse

App: follows every player and the ball, places them on the pitch, reads shirt numbers and finds the events.

About 0.75× the video's length.

3 · Report

You get:

a match overview, player profiles, event maps and the tracking drawn over the video.

4 · Review

You: correct a player's name.

App: re-runs the statistics with your correction.

The video is the only input. Everything else is worked out by the models.

Nine problems, and the answer to each

STAGES 1–3
Frames to tracks

1 · SAMPLE

Problem: A video is too heavy to read frame by frame

Answer: Frame stride, 12.5 frames a second

NO MODEL

2 · DETECT

Problem: Find every player and the ball in each frame

Answer: YOLOv8x detectors

PRETRAINED

3 · TRACK

Problem: Detections do not say who is who, and cuts break tracks

Answer: BoT-SORT + camera-cut rule

METHOD, NO TRAINING

STAGES 4–6
Pitch, teams and ball

4 · PITCH MAP

Problem: Pixels are not metres

Answer: HRNetV2-W48 + homography

PRETRAINED

5 · TEAMS

Problem: A detection does not say which side a player is on

Answer: K-means on shirt colour

METHOD, NO TRAINING

6 · BALL PATH

Problem: The ball is tiny and missed in many frames

Answer: Shortest-path search

HAND-WRITTEN RULES

STAGES 7–9
Names, events and totals

7 · IDENTIFY

Problem: Most shirt numbers cannot be read

Answer: ResNet-34 + PARSeq, CLIP identity model

FINE-TUNED / TRAINED BY US

8 · EVENTS

Problem: Tackles and shots leave little trace in positions

Answer: T-DEED spotter and event rules

FINE-TUNED / TRAINED BY US

9 · STATISTICS

Problem: Raw positions jitter, which inflates distance

Answer: Smoothing and step rules

HAND-WRITTEN RULES

Orange = fine-tuned or trained by us · green = pretrained · grey = hand-written or not learned · one RTX 3090, about 0.75× the video length

How do we find every player and the ball?

THE PROBLEM

A broadcast frame holds about 14 outfield players, goalkeepers, referees and a ball only a few pixels wide. Every later step depends on finding them first.

OUR ANSWER

USED AS RELEASED

YOLOv8x detectors

Data it uses: Video frames, 12.5 a second, resized to 1,280 px wide.

Trained on: Roboflow football images of players, goalkeepers, referees and the ball. Used as released.

HOW IT WORKS

1. Resize the frame to 1,280 px and pass it once through the network.
2. Predict a box and a class at three grid scales; merge overlapping boxes.
3. Keep boxes at 0.1 confidence or more. Later stages remove the false ones.

OUTCOME

89.7%

of visible player time detected

Weakest on close-ups: 63.6% for boxes over 150 px tall. About 14 outfield players are found in each frame.

Details: the model documentation, sections 2 and 5

How do we follow one player through the match?

THE PROBLEM

Detection looks at each frame alone, so nothing says two boxes show the same player. The camera also cuts about every 8 seconds, and a track must never carry on across a cut.

OUR ANSWER

METHOD, NO TRAINING

BoT-SORT tracker + camera-cut rule

Data it uses: Player boxes from the detector, and a colour histogram of each frame.

Trained on: Nothing. Fixed settings, no training data.

HOW IT WORKS

1. A Kalman filter predicts where each track goes next, with camera pans and zooms corrected.
2. New boxes are matched to tracks by overlap, confident boxes first.
3. A cut detector compares colour histograms: a distance above 0.35 is a cut.

OUTCOME

97.1%

of tracked time in single-player tracks

The cut detector found 115 of 133 labelled hard cuts. The demo match has 730 cuts, about one every 8 seconds.

Details: the model documentation, sections 2 and 5

How do we turn pixels into metres?

THE PROBLEM

Distance and speed need positions on the pitch, but the picture comes from a moving, zooming camera. Only a median 60% of broadcast time shows enough pitch lines to tell where the camera points.

OUR ANSWER

USED AS RELEASED

HRNetV2-W48 pitch network + homography

Data it uses: One video frame at 960 × 540 px.

Trained on: SoccerNet pitch images with labelled pitch points, from the original authors. Used as released.

HOW IT WORKS

1. The network finds 57 known pitch points: corners, box corners, centre circle, penalty spots.
2. RANSAC fits a homography from the matched points. Weak fits are rejected.
3. Fits are cleaned: outliers dropped, gaps up to 1.2 s filled, then smoothed.

OUTCOME

0.18 m

jitter after cleaning, down from 1.1 m

90th percentile, fixed points. The limit: only a median 60% of broadcast time shows enough pitch to calibrate.

Details: the model documentation, sections 2 and 5

Which team, which way, and where is the ball?

THE PROBLEM

A detection says “player”, not which side. Nothing says which way a team attacks, and the ball is missed in more than half of the frames where it is visible, with false candidates among the hits.

OUR ANSWER

HAND-WRITTEN

K-means, a defending rule and a path search

Data it uses: Torso colours of detected players, where the players stand, and ball candidates.

Trained on: Nothing. Hand-written, no training data.

HOW IT WORKS

1. Teams: two-cluster K-means on torso colour, with grass masked out.
2. Direction: defenders stay between ball and own goal, so the smaller mean position is the left goal.
3. Ball path: a shortest-path search through the ball candidates of each shot.

OUTCOME

97%

of detected players on the right team

Attack direction is right on all 8 halves that could be checked. The ball is located in a median 64% of live play.

Details: the model documentation, sections 2 and 5

Can we read the number on a shirt?

THE PROBLEM

A shirt number names the player, but in most thumbnails it cannot be read: the player faces the camera, is far away or is turned. Reading every one would produce confident wrong numbers.

OUR ANSWER

USED AS RELEASED

ResNet-34 legibility check + PARSeq reader

Data it uses: The torso crop of each tracked player, as a thumbnail.

Trained on: SoccerNet jersey-number images, from the original authors. Used as released.

HOW IT WORKS

1. Cut the torso from the player box and scale it to a thumbnail.
2. A ResNet-34 decides whether a number is readable at all.
3. PARSeq reads the digits. Readings from one track are combined into one answer.

OUTCOME

80.3%

of 1,211 tracklets right

Counting “not visible” as right. 76.8% of the tracklets with a readable number are read correctly.

Details: the model documentation, sections 2 and 5

How do we name a player whose number is hidden?

THE PROBLEM

A number can be read for only 38% of tracklets. Team-mates wear the same kit, and goalkeepers wear a kit that matches neither team, so colour alone does not separate players.

OUR ANSWER

FINE-TUNED BY US

CLIP ViT-B/16 identity model

Data it uses: 224 × 224 thumbnails of each tracked player.

Trained on: 48 FOOTPASS games: 6.0 million thumbnails of 1,485 players with true numbers, identities and goalkeeper flags.

HOW IT WORKS

1. One CLIP ViT-B/16 network takes a 224 × 224 thumbnail.
2. Two digit heads read the tens and units digits separately.
3. An appearance head groups one player’s thumbnails; a goalkeeper head flags keepers.

OUTCOME

98.3%

of tracklets matched to the right player

Goalkeepers found with 89.9% recall at 99.0% precision. Learning digits only from readable thumbnails lifted the right-number rate from 74% to 88%.

Details: the model documentation, sections 2 and 5

How do we keep a name across camera cuts?

THE PROBLEM

Every camera cut ends all tracks, so a player comes back as a new, unnamed track. Names have to be rebuilt after each cut from up to 24 thumbnails per track, without joining two players who are on screen together.

OUR ANSWER

FINE-TUNED BY US

Grouping and naming rules

Data it uses: Up to 24 thumbnails per track, read by the identity model.

Trained on: No extra training. Rules tuned on 49 SoccerNet tracking clips and checked on FOOTPASS.

HOW IT WORKS

1. Cut up to 24 thumbnails per track and combine their digit votes, weighted by confidence.
2. Group segments into players by appearance, never joining two seen together.
3. Name a group when one number holds at least 85% of its votes.

OUTCOME

95%

of the names given are right

The right player is named for 52.3% of visible time. The unnamed half is not lost: unnumbered players keep their statistics.

Details: the model documentation, sections 2 and 5

How do we find passes, shots and tackles?

THE PROBLEM

Some events leave little trace in positions: a tackle looks like any other touch, and the ball is often missed. The video itself shows what happened, so the model has to watch it.

OUR ANSWER

FINE-TUNED BY US

T-DEED action spotter

Data it uses: 8-second video clips: 100 frames at 12.5 a second, 796 × 448 px.

Trained on: SoccerNet ball-action games plus 48 FOOTPASS games, with action labels.

HOW IT WORKS

1. Watch 8-second clips: 100 frames at 12.5 a second.
2. An image network per frame and a temporal encoder-decoder place each action on the right frame.
3. Heads give one of 12 actions, its exact moment and the team. Overlapping clips are merged.

OUTCOME

0.628

mean average precision, unseen games

Pass 0.93, carry 0.90, shot 0.66, tackle 0.10. Fine-tuning raised it from 0.606.

Details: the model documentation, sections 2 and 5

How do we count passes and carries from positions?

THE PROBLEM

Positions give only dots and a ball path. Nothing says what counts as a touch, a pass or a sprint.

OUR ANSWER

HAND-WRITTEN RULES

Event rules over positions

Data it uses: pitch positions of every player and the ball path, in metres, 12.5 times a second.

Trained on: nothing. The numbers were set and checked on one annotated SoccerTrack match.

TOUCH

The ball’s speed changes by at least 4 m/s while a player is within 2 m of it.

POSSESSION

Touches by one player less than 2.5 s apart form one possession spell.

PASS

The gap between two spells in one camera view, at most 4 s. Same team: completed. Other team: intercepted.

CARRY

A spell in which the player moved the ball at least 2 m between his first and last touch.

SHOTS

Not a rule: the video model (T-DEED) spots shots and credits them to the player at the ball.

SPRINT

At least 3 steps in a row, about 1.2 s, at 25.2 km/h or faster.

Thresholds tuned on an annotated SoccerTrack match · full definitions in the model documentation

How far and how fast did each player run?

THE PROBLEM

Raw pitch positions jitter by decimetres, which at 12.5 samples a second looks like fast running. Summing them directly would overstate distance and invent sprints.

OUR ANSWER

HAND-WRITTEN

Smoothing and step rules

Data it uses: Each player’s pitch positions in metres, 12.5 a second.

Trained on: Nothing. Hand-written, no training data.

HOW IT WORKS

1. Smooth positions over 13 samples, about one second.
2. Measure steps every 0.4 s; discard steps over 11 m/s as errors; sum into distance.
3. A sprint is 3 steps in a row at 25.2 km/h or faster.

OUTCOME

Per player

distance, speed zones, sprints

Counts cover only the time a player was on screen, so team totals are fuller than player totals.

Details: the model documentation, sections 2 and 5

Four building blocks we did not train

USED AS RELEASED

YOLOv8x detectors

Does: Finds players, goalkeepers and referees. A second detector finds the ball.

Comes from: Football weights from the Roboflow sports repository (Ultralytics YOLOv8x).

We use it: Runs on every sampled frame, 12.5 times a second.

USED AS RELEASED

HRNetV2-W48 pitch keypoints

Does: Finds 57 points on the pitch lines in each camera view.

Comes from: No-Bells-Just-Whistles (Gutiérrez-Pérez and Agudo), trained on SoccerNet calibration data.

We use it: We fit a homography to turn pixels into metres, then clean the calibration over time.

USED AS RELEASED

ResNet-34 + PARSeq number reader

Does: Checks whether a shirt number is readable, then reads its digits.

Comes from: The SoccerNet jersey-number pipeline (Koshkina and Elder; PARSeq by Bautista and Atienza).

We use it: Applied to the torso of each player thumbnail. Only readable thumbnails vote.

METHOD, NO TRAINING

BoT-SORT tracker

Does: Links the boxes of one player from frame to frame.

Comes from: A standard tracker, run with camera-motion correction and no appearance model.

We use it: Our cut detector restarts every track at a camera cut.

Licences include AGPL-3.0, GPL-2.0 and CC BY-NC: fine for research, to be reviewed before commercial use

How we trained the models we own

FINE-TUNED

Identity model: CLIP ViT-B/16

Starts from: OpenAI CLIP ViT-B/16 weights.

Learns from: 48 FOOTPASS games: 6.0 million player thumbnails with true numbers, identities and goalkeeper flags.

Learns to: Read shirt digits, tell players apart and spot goalkeepers. Digits are learned only from readable thumbnails.

Checked on: 3 unseen FOOTPASS games, 13,610 tracklets.

Cost: 10,000 steps on one RTX 3090.

FINE-TUNED

Action spotter: T-DEED

Starts from: The SoccerNet 2025 T-DEED baseline.

Learns from: SoccerNet ball-action games plus 48 FOOTPASS games, some with partial labels.

Learns to: Find 12 kinds of action in the video, with extra weight on rare ones such as tackles and blocks.

Checked on: 2 unseen SoccerNet games and 3 unseen FOOTPASS games.

Cost: 3,000 steps, about 3 hours on one RTX 3090.

How well each part works

PART	JOB	MEASURED ON UNSEEN GAMES	RESULT	VERDICT
YOLOv8x detector	Finds players	Visible player time detected	89.7%	**Strong**
Kit-colour teams	Splits the two teams	Detected players on the right team	97%	**Strong**
BoT-SORT tracker	Follows players	Tracked time in single-player tracks	97.1%	**Strong**
ResNet-34 + PARSeq	Reads shirt numbers	Tracklets read right	80%	**Good**
CLIP identity model	Names players	Names right · visible time named	95% · 52%	**Partial**
T-DEED spotter	Finds events in video	Mean AP (pass 0.93, shot 0.66, tackle 0.10)	0.63	**Mixed**
Event rules	Passes from positions	Pass precision · recall	68% · 70%	**Good**
Ball detector	Finds the ball	Visible balls found	47%	**Weak**

IN SHORT Seeing and tracking players is strong. Naming players, tackles and the ball are the weak points, so team totals are more reliable than player totals.

All measured on matches the models never trained on · full charts and confusion matrices are in the backup slides

Row	Metric	Calculation	Test data	When it counts as right
YOLOv8x 89.7%	Recall / detection rate	Detected player-samples ÷ all true visible player-samples	3 unseen FOOTPASS games, 2.03M player-samples	Our box overlaps the true box with **IoU ≥ 0.5**, one-to-one matching (Hungarian)
Kit-colour 97%	Team accuracy	Detected players with the right team ÷ all detected players	Same 3 FOOTPASS games	K-means team = true team
BoT-SORT 97.1%	Track **purity**	For each track, the time spent on its main true player; summed and ÷ all tracked time	Same 3 FOOTPASS games	A track isn't mixing two different people
ResNet-34 + PARSeq 80%	Accuracy per tracklet	Tracklets read right ÷ 1,211 tracklets	SoccerNet Jersey 2023 test split	Exact number right, **or** "not visible" correctly said when no number shows
CLIP 95% · 52%	95% = **precision** of names; 52% = **coverage**	95% = right names ÷ all names given. 52% = visible player time with a name ÷ all visible player time	3 FOOTPASS games, 13,610 tracklets	Name matches the true shirt number and team
T-DEED 0.63	**Mean Average Precision (mAP)**	Area under the precision-recall curve per class, averaged over the 12 action classes	SoccerNet BAS test: 2 games (Reading–Fulham, Stoke–Huddersfield), 3,980 labelled actions	Right action type within **±1 second**
Event rules 68% · 70%	**Precision · recall**	Precision = real ÷ all we called (744 ÷ 1,098). Recall = found ÷ all real (744 ÷ 1,069)	SoccerTrack v2, 1 match, **annotated positions** (tests the rules only, not vision)	Right type **and** right player within ±1s
Ball 47%	Recall at confidence threshold 0.1	Frames where our top ball is right ÷ frames where the ball is visible (6,766). Precision at that threshold is 72%	SoccerNet tracking test clips	Within **15 px** of the true ball

Scaling up: more matches, better models

SCALE

More matches

Each match is processed on its own, about 0.75× its length on one RTX 3090, so a 90-minute match takes about 70 minutes. Only broadcast video is needed, so any match on video can be added.

IMPROVE

Better models

More training data helped: moving the identity model from 19 to 48 games raised the right-name share from 48.9% to 53.0%. The ball is still the weakest link: it is found in under half of the frames where it is visible.

NEXT

Name more players

About half of visible player time is still unnamed. In a test with the true squads, a line-up raised the right-name share from 53.0% to 64.2%; more thumbnails and the naming panel also give names, and unnamed players keep their statistics.

POSSIBLE EXTENSION

Player grading and playing styles

A reviewer would grade players against a list of styles, and a model would estimate the rest. Nobody has graded players yet, so it needs labels before it can be tested. Our player statistics would be its input.

Estimates: 0.75× is the median processing ratio on one RTX 3090 · grading and styles are a possible extension, not a result